# RITIK KUMAR

5th-year, BS+MS (Research) Physics | Indian Institute of Science, Bengaluru

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## EDUCATION

**Indian Institute of Science**, Bengaluru, India. — Aug. 2022 – July 2027 (expected)
(BS+MS) in Physics — CGPA: 7.5/10.0

- PhD-level coursework via Joint Astronomy Programme (IISc, IIA, RRI, ISRO): Fundamental of Astrophysics, Radiative Processes, Galaxies & ISM, General Relativity & Cosmology, Astronomical Techniques, Fluid Mechanics & Plasma Physics
- Linear Algebra, Multi-variable Calculus, Probability & Statistics, Machine Learning, Matrix Theory, Stochastic models & Application, Mathematical Methods in Physics, Computational Physics.

## RESEARCH EXPERIENCE

**Spectral Analysis of Seyfert Galaxies** — Dec. 2024 – Jan. 2026
*Advisor:* ***Prof. C. S. Stalin****, IIA, Bengaluru*

Conducted this independent research project in collaboration with Prof. C. S. Stalin.

- Developed a Python pipeline to cross-match NLSy1/BLSy1 catalogs with NuSTAR observations (12  radius) using angular-separation and signal-to-noise criteria.
- Reduced NuSTAR data with NuSTARDAS (FPMA/FPMB), including calibration, background filtering, and barycentric corrections.
- Performed broadband spectral modeling in XSPEC using cutoffpl, nthcomp, pexrav, xillverCp, and relxill to constrain coronal temperature, high-energy cutoff, and reflection properties.
- Performed spectral and timing-variability analyses across multiple epochs, generating energy-resolved light curves and calculating fractional variability with uncertainties.
- Measured coronal temperatures across multiple epochs with well-constrained uncertainties, finding evidence for temporal stability in several sources.
- **Reported coronal temperature measurements for multiple sources and investigated coronal geometry, thickness, reflection fraction, and variability. In one source, multi-epoch observations revealed significant coronal-temperature variability.**
- ***Manuscript in preparation.***

**Tracing Outflows Across Different Gas Phases in AGN** — Sep. 2025 – Apr. 2026
*Bachelor's Thesis*
*Advisor:* ***Prof. C. S. Stalin****, IIA, Bengaluru*

- Systematically investigated ionized outflows in AGN by cross-correlating the Million Quasars (MilliQuas) catalog with archival XMM-Newton observations using a 5  matching radius.
- Investigated connections between different outflow phases and their characteristic spatial and physical scales using X-ray observations.
- Identified several AGN lacking detailed outflow studies and performed XMM-Newton spectral modelling, finding multiple outflow components with velocities of 0.06–0.33c "UFO-Forest" in several systems.
- Modeling methodology: Began with a simple power-law continuum, then incorporated reflection components to isolate residual features associated with outflows. After identifying line-like signatures, Gaussian absorption/emission models (gabs, zgauss) were applied for detection. Finally, physical modeling was performed using zxipcf and XSTAR grids for the final spectral analysis.
- Performed spectral and timing-variability analyses, including energy-resolved light curves and fractional-variability measurements with uncertainties.
- We have found the sources which are showing the warm absorption and UFO from Fe K line. And interesting thing is, we have found 2 to 3 source for the first time and these sources have soft x-ray ultrafast outflows like the paper by J N Reeves found in PDS 456.
- Developed custom XSTAR photoionisation grids to model soft X-ray absorption features and Fe K-line absorption features that could not be adequately described by the available zxipcf model.

- **Manuscript in preparation on a multi-phase outflow system exhibiting both soft X-ray and Fe K ultra-fast outflow signatures, with measured velocities of 0.22c and 0.32c, respectively, using custom XSTAR photoionisation grids.**

**Lead Researcher – Relativistic Reflection Modeling in AGN** Jan. 2026 – Aug. 2026
**Full Member, ECAP X-ray Research Group**
Collaboration with **Prof. Jörn Wilms, Dr. Thomas Dauser, , Alexey Nekrasov**
*Erlangen Centre for Astroparticle Physics* ***(ECAP)****, Friedrich-Alexander-Universität, Germany*

- Leading an independent research project on developing and testing new relativistic reflection (relxill_ring) models for AGN X-ray spectroscopy.
- **Granted full membership in the ECAP X-ray research group (Jan 2026)**, including:
  - Full SSH access to Remeis Observatory computing cluster for large-scale data analysis, Active membership in Sternwarte X-ray research mailing list, Regular participation in weekly ECAP X-ray group research meetings
- Led the observational analysis, including complete XMM-Newton and NuSTAR data-reduction pipelines, calibration, filtering, spectral extraction, broadband spectral modelling, and MCMC-based parameter estimation.
- Tested and validated new relativistic reflection model variants (extended corona, ring-like corona given by Nekrasov, A. D, et al. 2025), with real observational data, analyzing reflection signatures, disk-corona geometry, height of corona, and spin measurements through detailed XSPEC/ISIS modeling. Monte Carlo performed to check the parameter distribution.
- Found a boost parameter of approximately 1.2, indicating that the inferred coronal geometry deviates from a simple ring configuration.
- **Manuscript is in Preparation.**

**Master's Thesis: Constraining AGN Coronal Geometry using X-ray Polarimetry and Relativistic Reflection** Aug. 2026 – Present
Advisors: **Prof. Jörn Wilms** and **Dr. Thomas Dauser**, Friedrich–Alexander University Erlangen–Nürnberg (FAU), Germany

- Investigating the geometry of the X-ray corona in Active Galactic Nuclei (AGN) through a joint analysis of *IXPE*, *XMM-Newton*, and *NuSTAR* observations.
- Modeling relativistic X-ray reflection using the `relxill_ring` framework to test whether coronal geometries inferred from X-ray polarimetry are consistent with broadband X-ray reflection spectra.
- Performing a detailed study of NGC 4151 by combining X-ray polarimetry and relativistic reflection modeling to constrain the physical properties of the corona.
- Extending the `relxill` framework to investigate physically motivated extended coronal geometries and compare their observational signatures with existing models.

**ML Researcher – GRB Light Curve Reconstruction** Apr 2025 – July 2026
**National Astronomical Observatory of Japan (NAOJ)** **Advisor: Prof. Maria G. Dainotti**

- Developed and evaluated multiple deep learning models in PyTorch, including ReFANN, Siamese Neural Networks, and Deep Gaussian Processes, for uncertainty-aware reconstruction of irregularly sampled Gamma-Ray Burst X-ray light curves.
- **Siamese network:** shared encoder, four-component fusion (query, reference, difference, Hadamard product $\rightarrow$ 128-dim), MC Dropout uncertainty ($p$=0.1, Gal & Ghahramani 2016), physics-informed composite loss (Gaussian NLL + $\lambda$=$10^{-3}$ monotonicity penalty).
- **ReFANN pipeline:** 522/545 GRBs processed (98.2%), 62% mean parameter uncertainty reduction vs. 39% benchmark, test MSE 0.0196; 5-fold CV, bootstrap resampling, z-score normalisation in log-space.
- **Deep Gaussian Process (DGP):** Implemented a two-layer stacked GP (GPyTorch, RBF kernel, Cholesky variational strategy, 35/25 inducing points) applied to 521/545 GRBs; achieved 11.6%, 12.3%, and 15.9% uncertainty reduction in $\log T_a$, $\log F_a$, and $\alpha$ respectively with $\alpha$ outlier rate of 1.92% — competitive with BNN while requiring no Monte Carlo weight sampling.
- Implemented MC Dropout convergence analysis ($n_{\text{samples}}$=200) and ablation studies across embedding dims (16/32/64) to empirically validate design choices.
- Applied robust statistics: IQR/1.35-based $\sigma$ estimation, 3-$\sigma$ MAD outlier rejection, spatial-correlation bootstrap ($\sigma$=2.0), 95% CI reporting across 544 GRBs.
- **Published: Kaushal, Dainotti, Kumar et al.** (2026), JHEAp. Code publicly available: github.com/Ritikkumar07 Light-Curve-Reconstruction-2.

– **Second paper manuscript is in preparation.**

**Physics-Informed Neural Networks for ODEs/PDEs** Jan. 2025 – Apr. 2025
*Advisor:* ***Prof. Rishita Das****, IISc, Bengaluru*

- Implemented PINNs in PyTorch to capture discontinuities in shock tubes.
- Compared PINNs with traditional numerical methods (Runge-Kutta, Euler Method, central difference), which proved more reliable for handling discontinuities.
- Solved 1D/2D/3D Burgers' equations using different initial and boundary conditions, Poisson equation. We matched our results of PINNs with Numerical method and they were nearly the same.
- Solved the 1D Euler's equation using PINNs with initial Sod Shock Tube, and we captured partial shocks.
- Applied CFL stability criterion (CFL = 0.5) with 400 grid points for stable time-stepping in compressible flow simulations with shock wave propagation.

## PUBLICATIONS & PRESENTATIONS

### Journal Articles

- Kaushal, Aditi, M.G. Dainotti, **Ritik Kumar**, et al. (2025). *Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves.* **Journal of High Energy Astrophysics (JHEAp)**.
  DOI: 10.1016/j.jheap.2025.100519

### Manuscripts in Preparation

- **Ritik Kumar**, C. S. Stalin, and J. Hota, "Variation in the Temperature of the Corona in Seyfert-type AGN And disk geometry," 2026.
- **Ritik Kumar**, C. S. Stalin, and J. N. Reeves, A possible multi-phase outflows in a Luminous Quasar.
- Ayush, **Ritik Kumar**, M.G. Dainotti, et al. "Comparative Study of Machine Learning Methods for Gamma-Ray Burst Light Curve Reconstruction"

### Conference Presentations

- **XRISM International Conference 2025, Japan** — Poster Presentation
  *A comparative analysis of the coronal properties of Broad Line and Narrow Line Seyfert 1 galaxies.*
- **Astronomical Society of India, Conference 2026, Guwahati** — Poster Presentation
  *Nature of Coronae in Bright Active Galactic Nuclei and affecting the surrounding medium.*

## TECHNICAL SKILLS

**High-Energy Astrophysics:** Broadband X-ray spectral modelling, timing analysis, X-ray polarization, relativistic reflection, photoionized absorption, multi-epoch AGN spectroscopy, physical model fitting.
**X-ray Data Analysis:** XSPEC, ISIS, HEASoft, NuSTARDAS, SAS, FTOOLS; calibration, spectral extraction, response generation, background modelling, parameter estimation. **Data Pipelines:** Automated Python pipelines for NuSTAR/XMM-Newton data reduction, spectral and timing products, batch processing, reproducible multi-epoch analysis.
**Machine Learning:** PyTorch, GPyTorch, Git, Linux, Physics-Informed Neural Networks (PINNs), Deep Gaussian Processes, Siamese, ReFANN, uncertainty-aware deep learning, etc, time-series reconstruction, surrogate modelling for scientific datasets.
**Programming:** Python (NumPy, SciPy, Astropy, Pandas, Matplotlib), Bash scripting, parallel programming.
**Scientific Computing:** Numerical optimisation, Monte Carlo methods, Markov Chain Monte Carlo (MCMC), Bayesian parameter estimation, uncertainty propagation, statistical model comparison.
**Astronomical Software:** DS9, fv, TOPCAT, Astroquery, HEASARC archive, SDSS database access.

## WORKSHOPS

**Fast Radio Bursts (FRBs) Workshop** Oct. 6–17, 2025
International Centre for Theoretical Sciences (ICTS), India
**Galaxies and Related Topics: Honoring Prof. Biman Nath** Jun. 6–7, 2025
RRI, India

## ACHIEVEMENTS AND INTERNSHIPS

* JEE Advanced 2022 – All India Rank 3424 (Top 0.3 % of 1,60,000 candidates)
* 2025 Summer internship at **National Astronomical Observatory of Japan** (**NAOJ**) with **Prof. Maria Giovanna Dainotti.**
* Research collaboration with the **Indian Institute of Astrophysics (IIA), Bengaluru since December 2024.**